

Hugo Miguel Miranda Barros, MS

✉ hugobarros96@gmail.com

📄 [linkedin.com/in/hugo-barros-8934a7b5](https://www.linkedin.com/in/hugo-barros-8934a7b5)



Summary

AI Engineer with 5+ years bridging research and production, specialised in computer vision across 2D and 3D data. Background in bioengineering with hands-on experience across the full ML pipeline — from data to deployment. Growing focus on GenAI/LLMs through independent study and personal projects.

Education

MS, Bioengineering — Faculty of Engineering, University of Porto

2014 – 2020

Thesis: [Aquaculture Fish Quality Control Using Synthetic Data](#)

Skills

ML/AI	GenAI (LangGraph, OpenAI SDK), LLMs, CNNs, Transformers, RNNs, Reinforcement Learning, Diffusion, NeRFs, classical computer vision/image analysis . Supervised learning: classification, segmentation, regression, clustering. OCR (tesseract). Full ML pipeline from data processing to deployment .
Coding	Python (PyTorch, Sklearn, Pandas, Matplotlib, NumPy), C++ , SQL (basic), LaTeX
Workflow Tools	WandB (MLOps), Confluence , Jira , Docker , Git , Ray (distributed computing), Flyte , Pandera , Poetry , uv
Cloud	Azure and AWS essentials (auto-scaling, distributed computing)
Robotics	ROS2 , Foxglove (visualisation), Simulators: Gazebo, HoloOcean
Sensors/Data	Camera, IMU, DVL, SSS, Multibeam sonar, LiDAR, hydro, microphone; data types: images, text, point cloud, temporal signals
Languages	Portuguese (native), English (fluent), Spanish (essential)

Professional Experience

AI Researcher, Ocean Infinity

Aug 2020 – Present

- **Product Owner** for two projects from inception to Proof of Concept: quarterly planning, stakeholder engagement, and task distribution:
 - Bayesian target search using **sonar** sensors: probability-based optimisation with a custom-built 2D simulation environment;
 - Real-time **alignment aid tool**: segmentation of underwater drill bit components to detect misalignment;
- **Classification and eventing** of underwater assets using CNNs and Transformers, including temporal aggregation of single detections;
- **Inspection**: segmentation and detection of assets, achieving **over 80% F1 score**, with complementary **OCR** (tesseract) for asset identification;
- **Auto-annotation** product for object detection in **side-scan sonar images**, reducing annotation time **from weeks to hours**;
- **Point cloud** noise reduction and shape completion using networks such as PointNet;
- Research into monocular and stereo **SLAM** using **NeRFs** and **Gaussian Splatting** for real-time navigation and 3D reconstruction; C++ integration via **ROS2** and FoxGlove;

- End-to-end pipeline ownership: data analysis, processing, model training and tuning, through to **deployment, user trials, and feedback loop**.

Research Intern & MSc Thesis, Abyssal

Feb – Jul 2020

- **CNN prediction models** for monocular **depth estimation and 2D/3D segmentation**, using **GANs** for domain transfer from synthetic to real data. Preceded by an internship on CNN-based depth regression using simulated data.

Research Intern, INESC TEC

Jan – Jun 2019

- Follicle detection and segmentation in 3D ovarian ultrasound images: comparative study of **classical methods** (Otsu, Canny, watershed, active contours) against a **CNN-based** approach.

Miscellaneous

Awards & Certifications

- **Merit Student Grant Award**, Calouste Gulbenkian Foundation (2014)
- [AWS Cloud Practitioner Essentials](#) (2020)
- [15 Days of SQL: The Complete SQL Masterclass](#), Udemy (2025)

Extra Activities

- [Basic Life Support and Defibrillator Certified](#); Robotics Professor and Karate Instructor.
- Former member of BEST-Porto, European Youth Parliament, and VP of a Local Youth Political Party — built organization, communication, and leadership skills.